High-Frequency Radio Observations of Galaxy Cluster Minihalos

by

Benjamin Colquhoun

A thesis submitted to the Victoria University of Wellington in fulfillment of the requirements for the degree of Master of Science in Physics.

Te Herenga Waka - Victoria University of Wellington

2026

Abstract

Diffuse radio sources have been found within the cool cores of a growing number of relaxed galaxy clusters. These faint "minihalos" are a result of synchrotron-emitting, highly relativistic electrons. With extents over several hundred kiloparsec, they are far beyond the expected extent of a single source of relativistic electrons. The processes of *in-situ* turbulent reacceleration and hadronic production have been proposed to explain this phenomenon. Spectral steepening is expected at high radio frequencies for the reacceleration scenario, however, the majority of previous studies have been performed at lower radio frequencies (144 MHz to 1.4 GHz) where this is difficult to detect.

To investigate potential steepening, in this thesis I calibrate and image Very Large Array observations at 10 GHz of 12 clusters hosting minihalos. This represents the first steps towards a high frequency survey of minihalos. Source-subtracted measurements of minihalo flux are performed for 4 clusters, enabling comparison with prior observations at lower frequencies. Steepening was not observed for RXJ1720+2638, with a spectral index $\alpha = 1.07 \pm 0.05$, suggesting a hadronic origin for the minihalo. The remainder of the sample requires more extensive analysis.

Challenges encountered during the calibration and imaging are discussed, and recommendations for future analysis of high frequency minihalo studies are presented.

Acknowledgments

People who helped me during my Thesis: Yvette, CASA people, sibling, raapoi.

Artificial intelligence tools, ChatGPT (OpenAI, versions 4o, 5.1, and 5.2), were utilised in a supportive capacity over the course of this thesis. Assistance was provided in generation of code templates, plot styling, background research and concept clarification, and proofreading for understanding and flow. These tools were not used for any analysis of data, formation of conclusions, or writing sections of the thesis. All scientific results, conclusions, and final edits are the author's own work.

Contents

Abstract ii

Acknowledgements iii

Contents iv

1 Introduction 1

1.1 Observations of Galaxy Clusters . . . 3

1.1.1 Optical . . . 3

1.1.2 X-ray . . . 3

1.1.3 SZ . . . 4

1.1.4 Radio . . . 4

1.1.5 Gamma . . . 5

1.2 Diffuse Radio Emission . . . 6

1.2.1 Production . . . 6

1.2.2 Minihalo detections . . . 7

1.3 This Thesis . . . 8

1.3.1 Aims . . . 8

1.3.2 Work . . . 8

1.3.3 Cosmology . . . 9

1.3.4 Layout . . . 9

2 Radio Halos 11

2.1 Synchrotron emission . . . 11

2.2 Lifetime of synchrotron emission . . . 13

2.3 Production mechanisms . . . 15

2.3.1 Turbulent reacceleration . . . 15

2.3.2 Observables of turbulent reacceleration . . . 17

2.3.3 Hadronic Production . . . 19

2.3.4 Observables of hadronic production 20
2.4 Previous minihalo studies 21
2.5 Interferometry 23

3 Calibration **25**
3.1 Our observations 25
3.2 How to calibrate a dataset 27
3.3 Flagging data 27
3.4 Calibration pipeline 27
3.5 Special mentions 27
3.6 Imaging 27
3.6.1 Self calibration 28
3.6.2 Final cleaning 28

4 Analysis Tools **29**
4.1 Final datasets 29
4.2 Calibration Error 29
4.3 Calculating the minihalo flux 30

5 Results **31**

6 Discussion **33**

7 Conclusion **35**

Bibliography **37**

Chapter 1

Introduction

When looking up into the sky on a clear night, those without an interest in astrophysics might imagine many of the points in the sky as galaxies just like our own, floating alone in the deep void of space. Although true to an extent, it doesn't paint an accurate picture of the cosmos. In reality, many of these are associated with galaxy clusters, massive structures of tens to thousands of individual galaxies that are gravitationally bound into a single system (somewhat analagous to the planets in our solar system, but on a vastly different scale).[1] Galaxy clusters are the largest gravitationally bound structures in the universe, with total masses of order 10^{14}–10^{15} $M_\odot$ and diameters larger than 1.5 million parsec.[2]

One might naturally expect that the majority of the cluster mass originates from the innumerable galaxies held within, however, only a few percent of the total mass comes from this baryonic matter[3] (such as stars, planets, etc). The remainder of the baryonic mass comes from a hot diffuse gas (10^7–10^8 K) spread throughout the cluster called the intracluster medium (ICM). This dilute gas ($\sim 10^3$ particles m^{-3}) is held together by the gravitational pull of the cluster, with the highest density in the core and sparsest at the edges (Meekins et al., 1971; Serlemitsos et al., 1977). Even then, this amount of visible baryonic matter only comprises 15% of the gravitational mass calculated from the dynamics of the cluster, a huge discrepancy that was a key piece of evidence for the existence of dark matter. Indeed, the rough consensus from decades of study on galaxy clusters has $\sim 3\%$ of the mass budget taken by galaxies, $\sim 12\%$ from the ICM, and a staggering $\sim 85\%$ from unobservable dark matter (Blumenthal et al., 1984; Vikhlinin et al., 2006; Pratt et al., 2019).

Between these clusters, long filaments of galaxies span across space, with their own warm-hot intergalactic medium (WHIM) mimicking the role of the ICM. The WHIM is cooler and

[1] Smaller clusters with only tens of galaxies are often known as groups.

[2] For those unaccustomed to astrophysical units, $M_\odot$ is the solar mass and defined as $1\ M_\odot \approx 10^{30}$ kg. A parsec is defined as $1\ \text{pc} \approx 3 \times 10^{16}$ m. These unusual units are far more suitable at the scale of the universe.

[3] Astronomers use the term 'baryonic' to broadly refer to all matter that is not 'dark' matter, rather than the proper particle physics definition.

more diffuse than the ICM, with densities $\leq 10^2$ particles m^{-3} and temperatures 10^5–10^7 K. These filaments are far larger than clusters, though not gravitationally bound, stretching up to hundreds of Mpc to create a cosmic web throughout the universe with galaxy clusters located at intersections between filaments (Böhringer et al., 2025). This cosmic web is the largest structure in the universe, a vast network of dark matter that we can trace via the observable galaxies and clusters (Bond et al., 1996; Pimbblet, 2005) with ongoing work in detecting and understanding these filaments observationally (Tudorache et al., 2025). Clusters are known to form in part due to accretion of the WHIM at intersections of the cosmic web, as well as by mergers between smaller clusters (Press and Schechter, 1974; Voit, 2005; Kravtsov and Borgani, 2012).

An important distinction should be made when analysing clusters, between "disturbed" clusters undergoing merger events or still in a turbulent state from a recent merger, and "relaxed" clusters which have settled into an equilibrium state. Relaxed clusters favour theoretical studies due to their greater spherical symmetries, fewer substructures, and therefore, simpler modelling (Mantz et al., 2016). Many of these relaxed clusters also feature a "cool core", where the dense central region of the cluster exhibits a lower temperature than the outer regions. The high density of the ICM in the core causes a large outflux of X-ray emission. The resulting cooling time may then be lower than the expansion rate of the universe (known as Hubble time), causing the core temperature to drop and an inwards flow of gas to develop (Peterson and Fabian, 2006; Hudson et al., 2010). Although cool cores are observed, there is little evidence for the predicted inward cooling flow, suggesting the existence of a counteracting heating mechanism. The most widely supported theory involves active galactic nuclei (AGN), supermassive black holes at the centre of some galaxies. AGN consume massive amounts of matter and produce radio synchrotron emission that forms jets and lobes, which have been observed interacting with the ICM to distribute energy in "radio-mode" feedback (Fabian, 2012). The exact method of this heating is still disputed and undergoing research.

Within the high mass subset of relaxed cool core clusters ($M_{500} > 6\times10^{14}$ $M_\odot$)[4] recent works have shown that 80% contain radio minihalos (Giacintucci et al., 2017). Less common in lower mass clusters, and unobserved within non-cool core clusters, minihalos are diffuse radio sources with currently unknown origins. This radio emission is localised within the cool core, with radii from 50 kpc to $0.2R_{500}$.[5] This emission cannot be directly formed by radio sources such as AGN, as the spatial extent of these halos is well beyond the expected extent for particles sourced from an AGN (e.g. Bagchi et al., 2002; Giacintucci et al., 2014b). Understanding the origins and physics behind minihalos is an ongoing process, with viable production methods proposed, but a lack of conclusive observational evidence to differentiate between them. This thesis will provide

[4]The value M_{500} refers to the total mass within a radius of R_{500}, defined by the distance at which the clusters average enclosed density is 500 times the critical density of the universe at the cluster redshift. The critical density is a cosmological quantity that has varied over the lifetime of the universe.

[5]The radius $0.2R_{500}$ is a standard metric to separate the core and the outer ICM. Within this radius, the cooling, AGN feedback, and similar processes become significant to the behaviour of the ICM.

new observational data and a generalised process to provide further evidence to differentiate between the proposed processes.

1.1 Observations of Galaxy Clusters

Different processes within the cluster, and more specifically ICM, provide information about the physical environment. This radiation is emitted across the electromagnetic spectrum and we design telescopes to capture and relay different bands in order to identify different components of what we observe. The question then becomes, how do observations at different frequency bands inform us to the structure and processes occuring within a cluster?

1.1.1 Optical

The vast majority of optical light from clusters comes from stars within the component galaxies. Although incredibly informing about the stars and galaxies themselves, in the context of minihalos, optical light has limited use. Optical observations are useful in determining the component galaxies of a cluster, alongside investigating the radial velocities of galaxies. Since the radial velocity is dependent on the gravitational potential of the cluster, the distribution allows us to estimate the clusters mass (Voit, 2005). This analysis is difficult for more distant clusters and disturbed clusters, as the dynamical state will affect the radial velocities and skew the resulting mass.

1.1.2 X-ray

The hot gas of the intracluster medium forms an optically thin plasma, which has a degree of ionisation that depends on the temperature and density. We observe an X-ray spectrum from this gas with both continuum and line characteristics, and photon energies between 0.1 keV and 100 keV.[6] The energy of the photons is dependent on the temperature of the ICM, resulting in most emissions being in the narrower range of 1 to 10 keV (Schneider, 2014).

Line emission occurs due to electron transitions between orbitals within an atom. The temperature and density of the ICM changes the measured rate of this line emission, however, this becomes limited at higher temperatures as the ionisation fraction increases. The distribution of line emissions can also be an indicator of abundance of elements such as C, N, Ne, Mg, Si, S, Ar, Ca, and Fe. This is particularly evident at lower temperatures ($\lesssim 2$ keV) where the total emissions can be dominated by line emission (Voit, 2005; Schneider, 2014).

Continuum radiation is due to brehmsstrahlung, or free-free radiation, which is where electrons are deflected via Coulomb interaction with nearby nuclei. This deflection is an

[6] X-ray emission is historically measured in kilo-electron volts (energy) rather than frequency or wavelength.

acceleration which induces a change in kinetic energy that is emitted as a photon. The spectral energy distribution of these emissions is related to the kinetic energy of the particles, and therefore, the temperature T of the ICM. The scale of emissions is similarly impacted by the electron number density n_e, which traces the gas density of the ICM. This gives rise to the following relationship to total continuum emissivity ϵ_c (Schneider, 2014).

$$\epsilon_c \propto T^{\frac{1}{2}} n_e^2 \tag{1.1}$$

X-ray spectroscopy is a powerful tool to characterise the ICM and a clusters dynamical state. In general, telescopes trade off between high spatial or spectral resolutions, i.e. between detecting exactly where a photon originated from or an accurate measurement of the photon energy. This becomes somewhat of a constraint when observing galaxy clusters as we require reasonable spatial and spectral resolutions for these large objects. One common instrument is the space telescope *Chandra X-ray Observatory* launched in 1999 . Many studies of minihalos use *Chandra* observations to map the structure of the cool core and investigate potential relationships to the radio emission (e.g. Giacintucci et al., 2024). However, due to the aging detectors and contaminant build up on *Chandra*, the sensitivity of the telescope is decreasing. This is particularly bad at low energies (< 2 keV), where sensitivity is below 10% of where it was two decades ago (Grant et al., 2024). A new telescope *XRISM* was launched in late 2023 and was designed as a successor to *Chandra*. Over the last year researchers have begun to utilise the greater resolution to more deeply probe the dynamics of the ICM in some clusters (Sarkar et al., 2025; XRISM Collaboration et al., 2025). Although not the focus of this thesis, X-ray observations provide key ancillary data in investigating the processes involved.

Cite/footnote telescopes

1.1.3 SZ

Actually write this you idiot

1.1.4 Radio

Radio observations cover several orders of magnitude in wavelength, with frequencies from 10 MHz to over 100 GHz. This massive range comes with several difficulties for astronomers, with the primary one being the angular resolution of the telescope. Angular resolution θ defines the smallest area that can be resolved given an aperture and the observing wavelength.[7] Equation 1.2 gives the formula for angular resolution in arcseconds, or $\frac{1}{3600}$th of a degree, for an aperture of diameter D and observing wavelength λ.[8]

$$\theta = 2.5 \times 10^5 \frac{\lambda}{D} \tag{1.2}$$

Radio waves range from 3 cm to 3 km, orders of magnitude larger than optical ($\sim 1\mu$m) and X-ray (0.01−10 nm). It is then clear that to resolve the same object at radio frequencies as X-ray

[7] The angular resolution of a human eye (with 20/20 vision) is 1 arcminute. *See note 7*

[8] Arcseconds (”), arcminutes ($1' = 60$”), and degrees are common units in astrophysics as most angles are measured with this unit.

or optical, vastly larger telescopes would be necessary. In order to observe over this range several types of telescopes have been developed, including single dish telescopes, interferometer arrays, and very long baseline interferometer arrays (VLBI).

The size of a single dish antenna is limited by both practical construction and from scientific usability. The largest dish is the *Five-hundred-meter Aperture Spherical Telescope* (FAST) located in Guizhou China, though of the 500 metre diameter only a 300 metre area is usable at a time. Furthermore, as an immovable dish, it is limited in its view of the sky, with a maximum angle of 40 degrees from zenith. Several smaller 100 metre diameter dishes exist such as the *Green Bank Telescope* in West Virginia United States, and the *Effelsburg Telescope* near Bonn, Germany. These dishes also have a limited view of the sky and variable amounts of steerability due to their size, alongside worsening resolution at increasing frequencies. These restrictions developed into interferometry, the use of an array of single dish telescopes in tandem to simulate a larger dish. The *Karl G. Jansky Very Large Array* (VLA) in New Mexico United States, is comprised of 28 fully steerable 25-metre radio telescopes mounted on railway tracks in a Y-shaped array, observing between 73 MHz and 50 GHz. The *MeerKAT* telescope in the Meerkat National Park of South Australia contains 64 13.5-metre telescopes and operates between 1000 MHz and 10 GHz. These arrays can simulate observations by dishes several kilometres in diameter, while simultaneously able to track objects across the entire observable sky. VLBI takes this concept even further by using radio telescopes across different continents, synchronised by atomic clocks, to simulate dishes comparable to the size of the Earth. This thesis uses observations taken by the VLA, and the physics of these telescopes will be discussed in more detail in Chapter 2.

Cite/footnote telescopes

Radio emission detected by these telescopes from clusters falls into the category of 'synchrotron' emission. This synchrotron emission is produced by particles travelling at relativistic velocities through a magnetic field, emitting photons at radio energies characterised by their velocity and the magnetic field strength (Feretti et al., 2012; Schneider, 2014). This is due to the Lorentz force ($\boldsymbol{F} = \boldsymbol{v} \times \boldsymbol{B}$) perpendicular to their direction of motion, which induces a helical path through the field, and therefore constant acceleration. This is analagous to brehmsstrahlung radiation, but occurs due to a magnetic interaction instead of electric, which is why it is somewhat rarely called 'magnetobrehmsstralung'. A more detailed discussion of the types of radio emission is given in Section 1.2. A review of the physical processes that give rise to this emission, and the processes spectral and spatial markers is given in Chapter 2.

1.1.5 Gamma

The last region of interest for minihalo research is in detecting gamma radiation, or gamma-rays. This is the high energy limit for astronomy, where telescopes are designed to detect photons with

energies in the MeV and GeV range.[9] These prove difficult to not only detect, due to the majority of rays being absorbed in the atmosphere, but also to accurately describe their origin (Schneider, 2014). The *Fermi Gamma-ray Space Telescope* was built specifically for this research and was launched in 2008 carrying the Large Area Telescope (LAT), with an energy range between 30 MeV and 300 GeV.

There are various sources for gamma-rays, such as Gamma Ray Bursts (GRB), which are extremely short and intense flashes. Many of these have been spatially located to coincide with bright AGNS called blazars, and rapidly spinning magnetised neutron stars called pulsars. Another source is from gamma-ray background radiation, which is expected via neutral pion decays resulting from proton-proton collisions. Attempts to detect gamma-ray emissions connected to a clusters ICM have been ongoing as they allow us to put limits on effectiveness of proposed mechanisms to create radio halos (e.g. Reimer et al., 2003; Ackermann et al., 2010; Zandanel and Ando, 2014). Though progress has been made for giant radio halos, in particular firm constraints for the Coma cluster (Brunetti et al., 2012), minihalos have a smaller spatial extent and are much fainter, with no conclusive studies. The role of gamma rays in determining the processes creating minihalos is detailed in Chapter 2.

1.2 Diffuse Radio Emission

It has long been known that some clusters exhibit diffuse radio synchrotron emission that cannot be directly associated with radio galaxies within the cluster. This synchrotron emission can take many spatial arrangements, with the primary categories being radio halos, radio relics, and revived AGN sources. Radio halos are extended sources that trace the ICM distribution and operate at the scales of giant radio halos, matching the full ICM extent, and minihalos that are contained within a clusters cool core. Relics instead reflect particles that have been reaccelerated to relativistic energies by shock waves travelling through the ICM, such as the violent energies released during merger events. Relics tend to be elongated and irregularly shaped, located at the clusters periphery instead of the centre. Revived AGN sources are radio plasma that has been emitted from an AGN and re-energised similarly to the radio relics. From this point, this thesis will focus on radio emission in the context of halos, and in particular, minihalos. Several types of diffuse radio emission are shown in Figure 1.1.

1.2.1 Production

A key question in relation to the radio halos is how they are produced. The relativistic particles producing this emission have a characteristic lifetime due to the energy losses as they travel through space, and therefore a limited diffusion length-scale in which we should be able to

[9] Like X-ray, gamma-rays are measured by their energy rather than frequency.

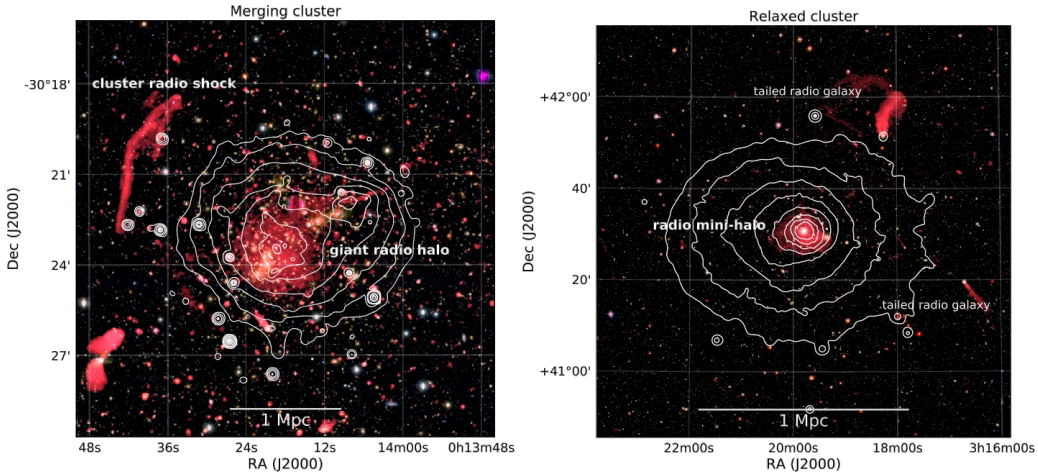

Figure 1.1: Left panel: Merging galaxy cluster A2744, VLA 1-4 GHz image (red) and Chandra X-ray contours (white) overlaid on Subaru BRz optical image. X-ray surface brightness contour are drawn proportional to $[1, 4, 16, 64, \ldots]$. The luminous giant radio halo and a cluster radio shock (relic) are labelled. Right panel: Relaxed cool core Perseus cluster, VLA 230-470 MHz image (red) and XMM-Newton X-ray contours (white) in the 0.4-1.3 keV band are overlaid on an optical image. X-ray contours have same spacing as the left image. The Perseus cluster hosts a radio mini-halo as well as two head-tail radio galaxies. (Figure 3 of (van Weeren et al., 2019))

observe them. Within a weak magnetic field ($\sim 10^{-7}$ G) these distances are about 10 pc, which even accounting for bulk motion of the ICM and more favourable diffusion environments are well below the Mpc scales observed in many clusters (Bagchi et al., 2002). Two primary theories have been brought forward to explain this phenomenon, one being turbulence in the ICM causing "reacceleration" of once relativistic particles (e.g. Brunetti et al., 2001; Gitti et al., 2002; ZuHone et al., 2013; Zandanel et al., 2014). The other is inelastic collisions between relativistic cosmic ray protons and thermal protons, termed "hadronic" or secondary production (e.g. Dennison, 1980; Dolag and Enßlin, 2000; Keshet and Loeb, 2010; ZuHone et al., 2015; Ignesti et al., 2020).

Both of these mechanisms produce similar observable features, which makes disentangling the true source much more challenging. Investigation into spatial extents and spectral features of giant radio halos has given the reacceleration theory more credence than hadronic production (Brunetti et al., 2012), however, the debate is still ongoing for minihalos. A detailed explanation of these processes is provided in Section 2.3. Due to their faint nature and smaller size, observations require high sensitivity and resolution in order to accurately capture an image.

1.2.2 Minihalo detections

Much effort has been devoted towards detection and validation of minihalos in order to build a comprehensive sample. Over the last two decades the number of detected minihalos has increased substantially, from 10 minihalos and minihalo candidates in 2011 (Feretti et al., 2012), to 22 in 2017 (Giacintucci et al., 2014b), and 40 in 2024 (Giacintucci et al., 2024).

One of the major issues with detection is in separating the minihalo emission from the much brighter sources embedded within and nearby it. Alongside the minihalo in the core is the brightest cluster galaxy (BCG), generally the most radio-loud AGN within the cluster. Minihalos typically have a surface brightness of $1 \sim 5$ μJy arcsec^{-2}, while the BCG can have a flux 1000 times higher.[10] The BCG must then be masked or subtracted away from an observation to reveal the minihalo, but this requires the observation to have good resolution of point sources as well as the larger spatial minihalos. This requirement makes current general sky surveys insufficient for detecting minihalos, with some studies suggesting that next generation radio telescopes could reveal thousands more minihalos due to improved capabilities (Gitti et al., 2018).

1.3 This Thesis

1.3.1 Aims

The primary goal of this thesis is to calibrate, image, and analyse new radio observations of 12 minihalos at 10 GHz. These observations were performed by the VLA in C and D configuration over two periods in early 2024 and early 2025. This data constitutes the first broad survey of minihalos at high radio frequencies.

This data will complement previous surveys at lower frequencies taken by the *Low-Frequency Array* (LOFAR) at 140 MHz and MeerKAT at 1.3 GHz (Riseley et al., 2022). Processing this data will allow measurements of the minihalo flux to investigate if steepening of the spectrum occurs at higher frequency, which will assist in differentiating between the reacceleration and hadronic scenarios.

1.3.2 Work

In pursuing these aims, I downloaded the raw interferometric data from the National Radio Astronomy Observatory (NRAO) Archive[11] and documented the calibration and imaging process. I then performed various analyses on the final images for each minihalo to evaluate the limits and usefulness of 10 GHz observations for characterising the production mechanisms of minihalos.

This work was done in a source controlled repository[12] so that the results are reproducible and documented. Due to the size of the datasets, (uncalibrated $\sim$ 1.4 TB compressed, calibrated

Add final size $\sim$ 2.5 TB, imaging $\sim$? TB), this is stored on the Storage fOr Learning And Research (SOLAR) large scale storage provided by Victoria University of Wellington.

[10] The Jansky (Jy) is a unit measurement of spectral flux density common in radio astronomy. It measures the rate of energy received per square meter per hertz, or 10^{-26}W m^{-2} Hz^{-1} in SI units.

[11] https://data.nrao.edu/portal/

[12] https://github.com/MoonlitMoo/Masters-2025

The repository is split into two primary sections, *Data Processing* and *Image Processing*. The first section follows the calibration of the raw data and separation from observation runs into cluster specific datasets, split by observation id. The following image processing includes the cleaning and creation of the images from the trimmed datasets and the associated analysis and is split by cluster.

1.3.3 Cosmology

Several physical values depend on the constants chosen for the cosmological model of the universe. This work assumes a flat ΛCDM model with Hubble's constant $H_0 = 70$ km s^{-1} Mpc^{-1}, fractional density of dark energy $\Omega_\Lambda = 0.7$, and fractional density of matter $\Omega_m = 0.3$.

1.3.4 Layout

The introduction has given a brief overview on several concepts necessary for understanding what minihalos are. The remainder of the thesis is organised as follows.

Chapter 2 provides deeper explanation of synchrotron radiation, the hadronic and turbulent reacceleration processes, and a summary of previous research into minihalos.

Chapter 3 introduces the collected data and explains the calibration and imaging process used.

Chapter 4 explains the tools and techniques used in the analysis of the final images.

Chapter 5 details the results found for each cluster, with the context of previous studies.

Chapter 6

Chapter 7

Chapter 2

Radio Halos

2.1 Synchrotron emission

Since the first discoveries of diffuse radio emission, much research has been devoted towards understanding the mechanisms behind the production of synchrotron-emitting electrons. These electrons are also known as 'cosmic ray' electrons (CRe) or non-thermal electrons to differentiate them from the thermal electrons of the ICM. The CRe have an individual energy $E = \gamma m_e c^2$ dependent on their Lorentz factor γ.

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \tag{2.1}$$

In the case of radio halos we can expect $\gamma \gtrsim 10^4$ which corresponds to velocities within 99.9999% the speed of light. Throughout their radiative lifetime CRe produce radio synchrotron emission with power dependent on the magnetic field B and their energy E. This is broadly a continuum radiation with the synchrotron power loss described by the following equation, where σ_T is the Tompson cross-section, and $U_B = B^2/(8\pi)$ is the energy density of the magnetic (Sarazin, 1999).

$$\left[-\frac{dE}{dt}\right]_{syn} = \frac{4}{3}\frac{\sigma_T}{m_e c}\gamma^2 U_B \tag{2.2}$$

They also interact with thermal protons in the ICM via inverse Compton (IC) scattering,[1] of which the dominant proton population originates from the Cosmic Microwave Background (CMB). The scattered photon has a boosted energy that is proportional to its original energy. In units of frequency this relation is:

$$\nu_p = \frac{3}{4}\gamma^2 \nu_{p,0} \tag{2.3}$$

For the CRe under consideration this results in a boosting factor $\sim 10^6$. For thermal photons sourced from the CMB, they move through the cluster with a temperature of T = 2.73 K, to

[1] Standard Compton scattering has ultrarelativistic photons interacting with thermal electrons, the opposite to what we are considering.

which the Planck function peaks at $\nu \approx 1.6 \times 10^{11}$ Hz. This interaction causes emission at $\nu \approx 10^{17}$ Hz or $\lambda \approx 1$ nm, corresponding to high energy X-rays (Sarazin, 1999; Govoni and Feretti, 2004). The energy losses due to IC scattering with CMB photons can be expressed as the CMB energy density $U_{CMB} = B_{CMB}^2/(8\pi)$, and an equivalent magnetic field strength $B_{CMB} = 3.25(1+z)^2$ [μG]. This magnetic field isn't real, but results in equivalent energy losses as if electrons were interacting with a real magnetic field within the cluster.

$$\left[-\frac{dE}{dt}\right]_{IC} = \frac{4}{3}\frac{\sigma_T}{m_e c}\gamma^2 U_{CMB} \tag{2.4}$$

These mechanisms are then highly dominated by the synchrotron and IC losses, which scale $\propto \gamma^2$, at high Lorentz factors and so can be neglected. The combined synchrotron and IC losses can then be written as a single loss equation where the magnetic fields are measured in μG. Further losses due to Coulomb and brehmsstrahlung interactions are possible, but scale with γ as follows (Sarazin, 1999).

$$\left[-\frac{dE}{dt}\right]_{Coul} \propto \ln\gamma, \qquad \left[-\frac{dE}{dt}\right]_{brehm} \propto \gamma \ln(\gamma)$$

These mechanisms are naturally highly dominated by the synchrotron and IC losses, which scale $\propto \gamma^2$, at high Lorentz factors and so can be neglected. The combined synchrotron and IC losses can then be written as a single loss equation where the magnetic fields are measured in μG.

$$-\frac{dE}{dt} = \frac{1}{6\pi}\frac{\sigma_T}{m_e c}\gamma^2\left(B^2 + B_{CMB}^2\right) \tag{2.5}$$

The standard assumptions regarding the population of CRe within the ICM are that it is homogeneous, isotropic, and follows a power-law energy distribution $n(E) = n_0 E^{-\delta}$. The total synchrotron emissivity ϵ_s at a given frequency ν is dependent on the CRe number density, the strength of the ICM magnetic field, and the spectral index of the CRe distribution.

$$\epsilon_s(\nu) \propto n_0 B^{(\delta+1)/2}\nu^{-(\delta-1)/2} \tag{2.6}$$

This can be rewritten in terms of $\alpha = (\delta - 1)/2$, the spectral index of the synchrotron emission.

$$\epsilon_s(\nu) \propto n_0 B^{\alpha+1}\nu^{-\alpha} \tag{2.7}$$

This relation shows a degeneracy between the magnetic field strength and the CRe number density, as such, the synchrotron radiation alone is insufficient to directly measure the magnetic field. Estimates of the magnetic field are generally found by minimising the total energy contributed by the relativistic electrons $(1+k)U_e$, where k is the ratio of energy of relativistic protons to the electrons, and the magnetic field U_B.

$$U_{tot} = (1+k)U_e + U_B \tag{2.8}$$

This occurs when the total energy of the relativistic particles is close to equilibrium with the total magnetic field energy $U_B = \frac{3}{4}(1+k)U_e$, and therefore $U_{min} = \frac{7}{3}U_B$, and is often referred to as the equipartition value (Govoni and Feretti, 2004). This allows the calculation of the minimum energy density u_{min} which can be used to derive the associated equipartition field B_{eq}.

$$B_{eq} = \left(\frac{24\pi}{7}u_{min}\right)^{\frac{1}{2}} \tag{2.9}$$

However, the IC emission is independent of the magnetic field, meaning that comparisons between these fluxes would allow direct measurement of the magnetic field strength (Govoni and Feretti, 2004; Wik et al., 2014). Unfortunately, detection of the X-ray IC component is difficult as low X-ray energies are dominanted by the thermal ICM emission, while upper energies lack spatial resolution in current instruments. An interesting topic in its own right, IC emission and numeric estimations of cluster magnetic fields are outside the scope of this thesis.

2.2 Lifetime of synchrotron emission

The energy losses of the CRe as they interact with the electromagnetic fields and particles in the ICM eventually result in them no longer being energetic enough to produce observable radio emission. Considering losses due to synchrotron and IC processes, the lifetime of a CRe τ can be given as the following function of the magnetic field in the ICM B, the equivalent magnetic field of the CMB B_{CMB} (both fields are measured in μG), the cluster redshift z, and the observed frequency of the particle ν (van Weeren et al., 2019).

$$\tau = 32\frac{B^{\frac{1}{2}}}{B^2 + B_{CMB}^2}\left[(1+z)\nu\right]^{-\frac{1}{2}} \quad \text{[Gyr]} \tag{2.10}$$

As the majority of observed minihalos have redshifts in the range $[0.1, 0.3]$[2] and magnetic fields within the ICM are estimated at $\lesssim \mu$G, the maximum expected lifetime of a CRe is about 10^8 years (Sarazin, 1999; Feretti et al., 2012; Gitti et al., 2018).

A basic check for the spatial extent for these particles, in absentia of external forces $d \approx c\tau$, finds a distance of $\sim 10^{16}$ m or 1 pc from the source. This is orders of magnitude below the extents of observed diffuse radio emission, so we then consider that the CRe are moving within the bulk motion and magnetic fields of the ICM. By assuming that there is a strong enough turbulence to promote diffusion of CRe, we can use the Bohm approximation $D = \frac{k_B T}{eB}$, which models the diffusion of plasma across a magnetic field. Bagchi et al. (2002) finds that the expected length-scale of this diffusion (at $z = 0.3$) is that of a random walk across its lifetime

[2] A handful minihalos are notably outside this range, with the Phoenix cluster at $z = 0.596$ (van Weeren et al., 2014), ACTJ0022-0036 at $z = 0.8$ (Knowles et al., 2019), and a recent candidate SpARCS1049+56 at $z = 1.709$ (HLavacek-Larrondo et al., 2025)

with a diffusion coefficient κ and length-scale L_{diff},[3]

$$\kappa = 3.31 \times 10^{21} \left(\frac{E}{\text{GeV}}\right)\left(\frac{B}{10^{-1}\mu\text{G}}\right) \text{ m s}^{-1} \tag{2.11}$$

$$L_{diff} = \sqrt{\kappa\tau} = 6.7 \left(\frac{B}{10^{-1}\mu\text{G}}\right)^{-\frac{1}{2}} \text{ pc} \tag{2.12}$$

Once again, we find that the expected diffusion and halo produced by a point source is measured in the 10s of parsecs, far below the kiloparsec and megaparsec halos observed. It then seems a reasonable conclusion that these particles are instead reaccelerated or produced *in-situ* by a process that is extended over the kiloparsec scale and long-lived across the particle lifetime.

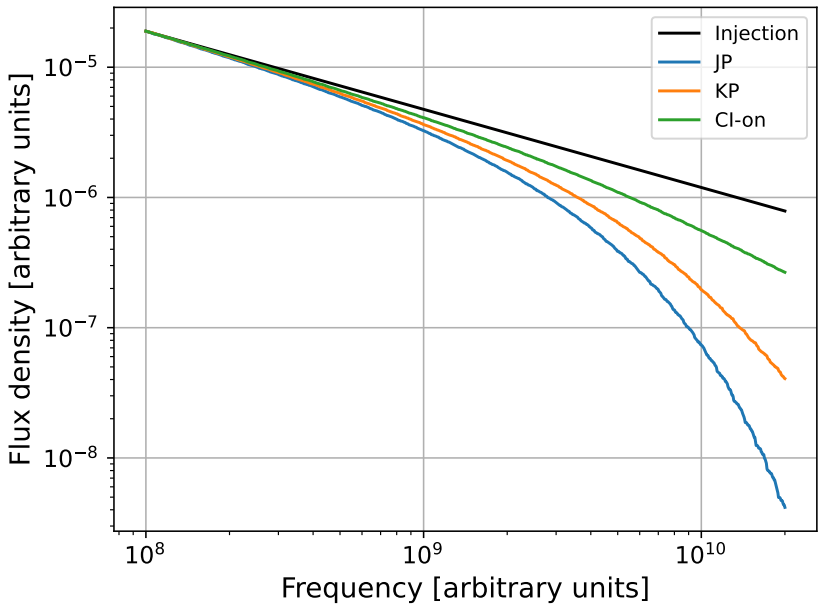

Figure 2.1: A representation of radio spectral shapes with $\alpha_{inj} = -0.6$. The injection spectrum depicts the shape prior to energy losses. The curves have been normalised to the injection spectrum at low wavelengths for better comparison.

The spectral shape of synchrotron emission is also affected by the lifetime of the particles, as higher energy particles have a shorter lifetimes ($\tau \propto \nu^{-\frac{1}{2}}$), causing the population of CRe change over time from the initial power-law. Several models have been developed to describe the resulting spectra, depending on the initial injection spectrum α_{inj}, break frequency, and any continuous injection after the initial state. The Jaffe-Perola (JP; Jaffe and Perola 1973) model assumes all CRe are created in a single burst, before aging with continuous scattering of electron pitch angles. This predicts that the pitch angle (between the electron velocity vector and local magnetic field vector) will be isotropic due to scattering. The Kardashev-Pacholczyk (KP;

[3]The length-scale diffusion is calculated as a random walk, scaling with the square root of time, rather than linearly.

Kardashev 1962, Pacholczyk 1970) model is similar to the JP model as it uses the same initial conditions, but evolves without pitch angle scattering. Continuous injection (CI) models describe sources that exist for longer than a single burst of acceleration, such as until current time (CI-on; Kardashev 1962), or for a fixed amount of time τ_{on} (CI-off; Komissarov and Gubanov 1994). An example of how these models differ in their shape is provided in Figure 2.1. The initial burst models show extreme steepening towards high frequency. In contrast, the CI model offsets the losses at high energy due to newly injected particles, but still has a lower level of steepening due to the presence of aged particles alongside newer ones.

The basic observables of synchrotron radiation can then be summarised in the following points.

- The synchrotron emissivity is dependent on the number density of the CR electrons and the strength of the magnetic field (Equation 2.7). Assuming a constant magnetic field strength, the emissivity follows a power-law $\epsilon_s(\nu) \propto \nu^{-\alpha}$ which is related to the CRe energy spectral index δ by $\alpha = (\delta - 1)/2$. Typical minihalo spectral indices are steep with $\alpha > 0.7$.[4]

- As the emitting particles lose energy the particle energy distribution changes, and therefore, the emitted radio spectrum changes. This appears as a steepening or curved spectrum at high frequencies, rather than a single power-law. The break frequency at which this occurs is related to the time since acceleration. This means that radio events that have stopped will show an aged spectrum with a characteristic break, i.e. radio relics. Ongoing events that are continually injecting new particles will not feature the same level of steepening.

2.3 Production mechanisms

With this understanding of the spectrum of CRe, we now consider the methods that can produce them. This can be broken down into primary models involving first and second order Fermi acceleration, and secondary models where the electrons are a byproduct or resulting from decay.

2.3.1 Turbulent reacceleration

This process takes a pre-existing lower energy CRe population of $\gamma \sim 100$ and boosts it to the factors observed $\gamma \gtrsim 1000$. This occurs via interactions with plasma waves and shocks within the ICM which can be caused by several factors, such as violent cluster mergers and gas sloshing within the core (Brunetti et al., 2001; Fujita et al., 2004; Mazzotta and Giacintucci, 2008).

Shocks within the ICM can cause first order Fermi acceleration (Fermi-I), also known as diffuse shock acceleration (DSA). As particles cross back and forward over the shock front, they can be reflected by magnetic instabilities upstream and downstream from the shock. Each

[4] A typical radio galaxy emits synchrotron radiation with a spectral index ~ 0.7.

crossing of the front increases the particle energy, raising trapped particles to relativistic levels before they escape. The energy gained via Fermi-I acceleration from a shock with velocity v_{wave} scales, compared to the speed of light c, as

$$\Delta E \propto \frac{v_{wave}}{c} E \tag{2.13}$$

Shocks are generally created due to merger activity in disturbed clusters, not relaxed clusters where minihalos are found. As such, it is not considered a major mechanism at play in the context of minihalos.

Gas sloshing within the cool core of relaxed clusters has been shown as a natural consequence of the larger motions of the ICM (Fujita et al., 2004). This turbulent environment naturally allows for low relativistic 'seed' electrons to undergo second order Fermi (Fermi-II) acceleration. This type of acceleration is stochastic as particles are scattered by the magnetic instabilities from the randomly moving plasma waves. Particles meeting these waves head-on are reflected with an increased energy, while particles in a head-tail collision are reflected with a lower energy. The chance for a head-on collision is higher than that of a head-tail collision allowing for a net gain. The mean energy gained is dependent on the speed of the waves and energy of the seed particle,

$$\Delta E \propto \left(\frac{v_{wave}}{c}\right)^2 E \tag{2.14}$$

The random nature of Fermi-II acceleration makes for an inefficient process compared to the aligned shock fronts of Fermi-I. Since $v_{wave} \ll c$, Fermi-II scales far more slowly than Fermi-I $(v_{wave}/c)^2 \ll v_{wave}/c$.

Brunetti and Lazarian (2007) show that reacceleration of relativistic particles moving through magnetohydrodynamic (MHD) turbulence are primarily accelerated by two Fermi-II type mechanisms.

The first mechanism is the damping of fast magnetosonic waves due to resonant transit time damping (TTD). The compressible and low-frequency plasma waves can resonate with the naturally gyrating particles at the zeroth order, $\omega = v_{\|} k_{\|}$, where ω is the frequency of the wave, $k_{\|}$ is the wavenumber, and $v_{\|}$ is the speed of the particle along the axis of the magnetic field. The random accelerations can then be captured as a random walk in momentum space, giving rise to a momentum-diffusion coefficient $D_{pp,TTD}$. ZuHone et al. (2013) simulate this process numerically in the case of minihalos and simplify this diffusion coefficient to be proportional to the average wavenumber $\langle k \rangle$, the turbulent velocity of the particle v_t,[5] and fraction of turbulent energy in compressible motion R_c.

$$D_{pp,TTD} \propto \langle k \rangle v_t^2 R_c p^2 \tag{2.15}$$

The second mechanism is through the non-resonant compression and expansion of the ICM. Adiabatic compression increases the particle momentum, while expansion reduces it. As with

[5]This is essentially the instantaneous velocity of the particle due to turbulence.

collisions, the particle is more likely to experience compression than expansion, leading to a net gain in energy. Here ZuHone et al. (2013) take the fast diffusion limit, where particles escape the eddies of compression before they re-expand. This non-resonant momentum-diffusion coefficient scales as

$$D_{pp,C} \propto \frac{V_l^2}{D} p^2 \tag{2.16}$$

where V_l is the compressive velocity at scale l, and $D \propto c l_{mfp}$ is the spatial diffusion rate for a mean free path of l_{mfp}.

The reacceleration efficiency is then given via $\chi = \frac{4D_{pp}}{p^2}$, where D_{pp} is the sum of the prior diffusion coefficients. The total rate of momentum increase for each particle can then be clearly written as the sum of two Fermi-II processes when compared to Equation 2.14.[6]

$$\frac{dp}{dt} = \chi p \tag{2.17}$$

$$\frac{dp}{dt} \propto \langle k \rangle R_c v_t^2 p + \frac{1}{D} V_l^2 p \tag{2.18}$$

One remaining loose thread in this explanation is the source of the seed electrons themselves. A possible explanation is that the seed population is provided from radio AGN and spread throughout the cluster core by the gas sloshing (Brunetti et al., 2001). The CRe from the central BCG lose energy and disappear out of our detectable range by the physics described earlier, however, they don't stop existing after this timescale. These 'relic' CRe are lower energy but still relativistic with $\gamma \sim 100$, radiating at radio frequencies that we are unable to detect. In this state they still continue to move and disperse through the core, creating the seeds for the large scale minihalos that we observe. The central BCG may not be the only source, other AGN within the cluster, galactic winds, supernovae, may all contribute relic CRe to be later reaccelerated. ZuHone et al. (2013) find that the sloshing should be sufficient to redistribute these particles to create a minihalo. Hadronic interactions as described in a later section could also contribute seed electrons.

2.3.2 Observables of turbulent reacceleration

With this understanding we can now explore the spectral and spatial features that we expect to observe from emission resulting from the reacceleration scenario. Turbulence is an inherently transient and intermittent phenomenon. This spatial variance means that when we sample a given region of space, we may instead be observing several 3D components superimposed on our 2D image. Areas with higher turbulent velocity inspire greater magnetic fields and greater levels of emission (ZuHone et al., 2013). The lack of a steady state and variable components means that we might expect variation in the flux brightness and spectral index across different locations of

[6]The factors of c have been absorbed into the constants, with the main focus being the squared scaling with velocity and linear with momentum

the minihalo (Riseley et al., 2022). However, due to the integration over the line of sight, these artefacts could be averaged and less visible in the 2D observation.

The particles must also trace the distribution of turbulence within the ICM, leading to a spatial bounding of the minihalo within the gas sloshing of the core. The boundary of the cluster core is often associated with cold fronts visible in X-ray data, allowing comparison between X-ray and radio images. Cold fronts denote the boundary between cooler denser gas clouds moving through a hotter and more diffuse surrounding gas. These are notably different to shock fronts, where the boundary separates a hot and dense downstream region that the shock has passed through, from the cooler and less dense upstream region the shock is moving towards (Markevitch and Vikhlinin, 2007). ZuHone et al. (2013) find that simulated turbulent re-acceleration is indeed bounded by the cold fronts (Figure 2.2), with a sharp steepening at the core radius. They also find that depending on the projection through the ICM, it may be difficult to locate the cold fronts in X-ray images due to the afore mentioned line of sight integration.

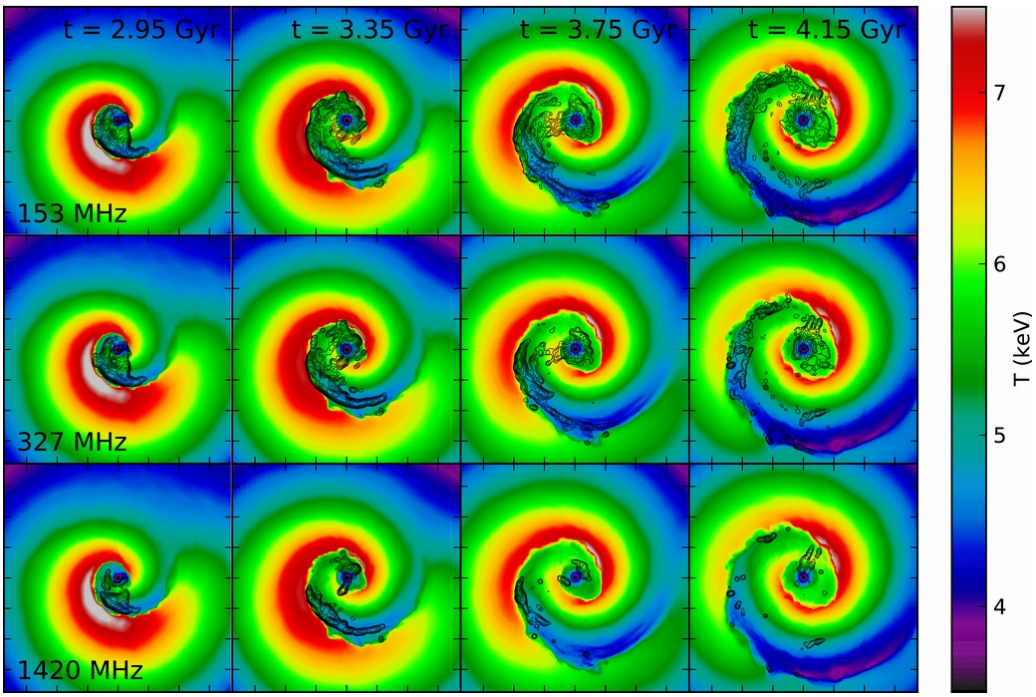

Figure 2.2: 2D projected gas temperature maps with overlaid radio contours at several simulation timesteps for frequencies 153, 327, and 1420 MHz. Radio emission contours begin at $(1.0, 0.5, 0.125) \times 10^{-3}$ mJy arcsec^{-2} and increase by a factor of 2. Image dimensions are 750×750 kpc, with tick marks every 100 kpc. (Figure 10 of ZuHone et al. (2013))

A spectral feature of the reacceleration model arises from the balancing of reacceleration and losses, creating a cutoff energy at which the population of CRe declines more sharply. This is indeed the logic behind the CI model visualised in Figure 2.1, with a steepening in the spectral index beyond the break frequency. ZuHone et al. (2013) note that as there is a range of magnetic field strengths within the core, this break frequency will change between CRe, causing a smearing of the break and a more gradual steepening towards high frequency.

2.3.3 Hadronic Production

For the secondary production we instead consider the protons within the ICM, rather than the electrons. We again separate them into a relativistic 'cosmic ray' proton (CRp) population and a thermal population and examine the resultant interactions. CRp have significantly longer radiative lifetimes than CRe, with $\tau_p \sim 10^{10}$ yrs, allowing them to travel up to 1 Mpc (Dennison, 1980). The spread out CRp (p_{CR}) then collide inelastically with thermal protons (p_{th}) to produce charged and neutral pions π

$$p_{CR} + p_{th} \rightarrow \pi^0 + \pi^+ + \pi^- \tag{2.19}$$

the charged pions then decay into (anti)muons $\mu^\pm$ and muon (anti)neutrinos $\nu_\mu(\tilde{\nu}_\mu)$, while the neutral pions decay into gamma rays γ

$$\pi^\pm \rightarrow \mu^\pm + \nu_\mu(\tilde{\nu}_\mu) \tag{2.20}$$

$$\pi^0 \rightarrow 2\gamma \tag{2.21}$$

the muons then decay into highly relativistic (positrons)electrons $e^\pm$ and electron (anti)neutrinos $\nu_e(\tilde{\nu}_e)$.

$$\mu^\pm \rightarrow e^\pm + \nu_e(\tilde{\nu}_e) + \tilde{\nu}_\mu(\nu_\mu) \tag{2.22}$$

From this chain reaction we end up with two observables, CRes and gamma rays. Importantly, it is these secondary CRe products that are predicted to produce the minihalo emission. The emission in this model then traces the population of CRp producing the CRe, rather than the turbulence of the ICM. Due to our assumption of the long radiative lifetimes, we expect a continuous injection of CRe from the hadronic process.

ZuHone et al. (2015) investigate the hadronic method via simulation and assume a power-law momentum distribution for the CRp, $n_p(p_p) = C_p p_p^{-\alpha_p}$ (where C_p is a normalisation constant), and estimate the CRp momentum distribution spectral index to be in the range $\alpha_p \sim 2.0$-2.5. They then derive a distribution for the produced CRe momentum p based upon the CRp momentum distribution to be

$$Q(p)dp = \frac{4}{3} 16^{1-\alpha_p} c n_{th} C_p \sigma_{pp}(16p) p^{-\alpha_p} dp \tag{2.23}$$

with n_{th} the density of thermal protons, $\sigma_{pp}(16p)$ the proton-proton inelastic collision cross section at 16 times the CRe momentum.[7] Ignoring the numerical terms we can also write

$$Q(p)dp \propto n_{th} \sigma_{pp} p^{-\alpha_p} dp \tag{2.24}$$

to clearly see that the efficiency of the hadronic method scales with the density of thermal protons, collision cross section, and as a power-law in momentum. By approximating σ_{pp} as constant[8]

[7] This is an approximation of the CRp momentum in the proton-proton collision, $p_p = 16p$.

[8] There is a weak dependence on the CRp and thermal proton energies, which ZuHone et al. (2015) account for, resulting in a minor flattening effect on the synchrotron spectral index.

we are able to find a steady state for the number density of CRe by equating the synchrotron and IC losses $\frac{dp}{dt} \propto -p^2$. This occurs when $\frac{d}{dp}(-p^2 n_e) + Q(p) = 0$, which following through by integrating for all particles $p \to \infty$,

$$-p^2 n_e = -\int_p^\infty Q(p)$$
$$p^2 n_e \propto \int_p^\infty p^{-\alpha_p}$$
$$n_e \propto \frac{p^{-\alpha_p+1}}{p^2}$$
$$n_e \propto p^{-(\alpha_p+1)}$$

This gives us a CRe density power-law with index $\delta = \alpha_p + 1$ and therefore a synchrotron spectral index of $\alpha = \frac{\alpha_p}{2}$. Using the expected values for α_p we then should expect $\alpha \sim$ 1-1.25 which is similar to that of observed minihalos. However, this assumes steady state injection which is a somewhat idealised case for the intermittent and turbulent core of the cluster.

2.3.4 Observables of hadronic production

This hadronic process predicts a flat spectral index extending into high frequencies, lacking the steepening observed in the reacceleration model. This also implies the spectral index will be constant throughout the spatial extent of the minihalo, rather than varying due to localised high amounts of turbulence. Keshet and Loeb (2010) show that changes in the magnetic field strength relative to B_{CMB} can induce changes in the emissivity. For regions of the magnetic field where $B < B_{CMB}$ they predict a radio suppression $\propto (B/B_{CMB})^2$. Assuming that the magnetic field inside a cluster has a radial steepening, there then exists a radius r_b where the magnetic field drops below B_{CMB} and a radial break may be observed in the radio profile. As such, the extent of the minihalo would be more constrained via the strength of the magnetic field, than the cold fronts of the cool core.

The simulation of hadronic production by ZuHone et al. (2015) find that the majority of particles have spectral indices close to that of steady state. As B decreases towards $B \ll B_{CMB}$ the spread of spectral indices converges tightly. The opposite happens at $B \gg B_{CMB}$ with broadening and a small skew towards steeper indices. They attribute the broadening to sampling regions with rapidly decreasing magnetic fields (flatter index) and rapidly increasing magnetic fields (steeper index). They also note that the spatial bounding of the hadronic emission is less constraining than that of the reacceleration model. A drop in the emission across the cold front occurs due to a drop in the magnetic field strength and falls $\propto B^2$, but they simultaneously observed emission extending outside of the cool core.

Importantly, ZuHone et al. (2015) do not find evidence of a spectral flattening at high frequencies, showing that a simple power-law relationship holds to beyond 10 GHz. They

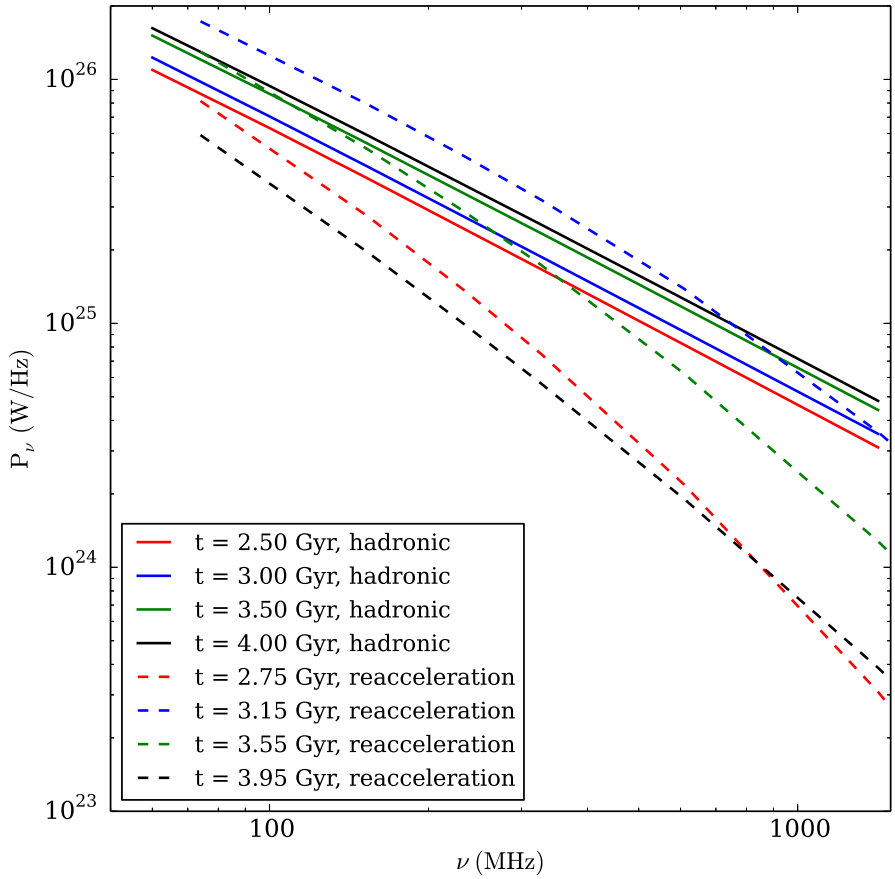

Figure 2.3: The total simulated radio spectrum within 300 kpc radius of the cluster centre, comparing the hadronic and reacceleration model at several timesteps. (Fig 8 of ZuHone et al. 2015)

compare the spectral distribution of the hadronic and reacceleration models and clearly show the difference in spectral shape (Figure 2.3).

A final clear distinction from the reacceleration model is the emission of gamma rays with the hadronic model. This implies that there will be a diffuse gamma emission aligned spatially with the minihalo. This would be a 'smoking gun' proof for the hadronic model, however as mentioned earlier, current instruments don't have the spatial resolution to accurately detect diffuse gamma emission at minihalo scales.

2.4 Previous minihalo studies

Minihalos are the most poorly understood class of diffuse radio emission, with few in-depth spectral studies to look at. For many minihalos we only have a couple of observations between 144 MHz and 1.4 GHz taken by the *Giant Metrewave Radio Telescope* (GMRT) and VLA (Giacintucci et al., 2014b, 2019). This lack of data means that for many minihalos the spectral index is calculated from a few points, if it can be at all. Furthermore, the lack of high frequency

data > 1.4 GHz makes detecting steepening from the reacceleration theory far more challenging, with both models resembling a power-law within this range (Figure 2.3).

The most in-depth spectral study at a range of frequencies has been performed on RXJ1720+2638 (Mazzotta and Giacintucci, 2008; Giacintucci et al., 2014a; Perrott et al., 2021). This revealed a tail to the minihalo, with a steeper spectrum than the central region. Early analysis showed a potential for steepening at 5 GHz, however, later observations at 13-18 GHz were consistent with a power-law index $\alpha = 1.0$ (Perrott et al., 2021). These spectral features lend impetus towards the hadronic scenario for the core, while also suggesting that reacceleration might play a role within the tail. The tail of the minihalo also appears to be constrained by the cold fronts of the core. More detailed analysis for this cluster is given in Section **??** , which includes images produced as part of this thesis.

Fix reference once we get there

The recent LOFAR-meets-MeerKAT series (Riseley et al., 2022, 2023, 2024) aims to perform a more uniform sample of minihalo observations to investigate the method of CRe production. These observations are performed at 145 MHz to 1.7 GHz with the afore mentioned LOFAR and MeerKAT telescopes and used alongside archival *Chandra* X-ray data. These detailed analyses of MS1455+2232, A1413, and A2142 have revealed that the minihalo morphology generally contains multiple components, a single steep-spectrum power-law, and spatially homogeneous spectral index. They find that both scenarios can explain several observables, but not all, and suggest that both mechanisms might be at play. MS1455+2232 and A1413 are discussed in more detail, alongside images produced during this thesis, in sections **??** and **??** respectively.

Fix reference

Another point of interest is in a possible connection to giant radio halos. Although minihalos are found exclusively within cool-core clusters and giant radio halos are found almost entirely within disturbed clusters, they may have a physical relation to each other. The general consensus is that giant halos are formed by reacceleration due to merger-induced turbulence, due to a lack of gamma emission at the larger cluster scale (Brunetti et al., 2012; Brunetti et al., 2017). These two phenomena may have a transitional state where a minihalo switches into a giant halo due to a merger, and the inverse as the cluster relaxes (Brunetti and Jones, 2014). An alternative scenario could be that mini-halos are produced via the hadronic process, which shifts towards turbulent reacceleration at larger scales (Zandanel et al., 2014).

The cluster A2142 shows some evidence towards these theories, with a large scale 2 Mpc halo comprised of two parts (Venturi et al., 2017). This cluster is undergoing a minor merger and has only small disturbances in the ICM. The brighter inner core component has a spectral index $\alpha_i = 1.33 \pm 0.08$, which appears to be produced similarly to that of a minihalo. The outer component is fainter and somewhat steeper with $\alpha_o \approx 1.5$, suggested to be caused by large scale turbulence within the ICM. Clusters may also show evidence of a minihalo and giant halo existing simultaneously, with RXJ1720+2638 and PSZ139.61+24.20 both exhibiting large scale steep spectrum emission outside the core region (Savini et al., 2018, 2019). Physical relations

to giant radio halos are useful in our understanding of how mergers change the behaviour of a cluster, but are outside the scope of this thesis.

2.5 Interferometry

To better understand the reasoning behind the calibration and imaging processes, it is useful to first cover the basics of interferometry. Interferometer arrays consist of individual antennas that measure the amplitude and phase of incoming electromagnetic waves. These signals are then combined with each other to produce an interference pattern. The simplest example for this is an array of two antennas. Since each antenna has a small difference in distance from the source, there is a geometric time delay τ_g between each receiving a signal. By combining the two individual signals, the interference pattern is created due to the phase shift caused by the time delay. Given a cosinusoidal wave travelling at the speed of light c, from a source in the far field at $\boldsymbol{s}$, and the baseline distance between the two antennas $\boldsymbol{b}$, the geometric delay is $\tau_g = \boldsymbol{b} \cdot \boldsymbol{s}/c$. The measured interference pattern is then the product of the individual fields $V_1 = E\cos(\omega t)$ and $V_2 = E\cos(\omega(t-\tau_g))$, which can be written in the form $\frac{E^2}{2}\left[\cos(\omega\tau_g) + \cos(2\omega t - \omega\tau_g)\right]$ (Perley, 2018). The second term varies much faster than the first and can be averaged out, and the prefactor can be shortened to $P = \frac{E^2}{2}$. This gives the final cosine response as

$$R_C = P\cos(\omega\tau_g) \tag{2.25}$$

or written in terms of the signal wavelength $\lambda = 2\pi c/\omega$ and the baseline,

$$R_C = P\cos\left(2\pi\frac{\boldsymbol{b}\cdot\boldsymbol{s}}{\lambda}\right) \tag{2.26}$$

The fringes of the interference pattern then sample the sky, with longer baselines corresponding to small spatial features, and small baselines mapping large spatial features. For the sky brightness map $I_\nu(\boldsymbol{s})$ at the frequency ν, the interferometer response is the spatial integral[9] of the sky modified by the cosinusoidal fringe pattern.

$$R_C = \iint I_\nu(\boldsymbol{s})\cos\left(2\pi\nu\frac{\boldsymbol{b}\cdot\boldsymbol{s}}{c}\right)d\Omega \tag{2.27}$$

This response function is even due to the cosine term, resulting in any odd functions described by the intensity map being averaged out. By modifying the phase of one of the antennas by 90 degrees, $V_2 = E\sin(\omega(t-\tau_g))$, and applying the same processes as before, an odd response function is created $R_S = P\sin(\omega\tau_g)$. The two responses R_C and R_S can then be used to define the complex visibility function in terms of $\boldsymbol{u} = \boldsymbol{b}/\lambda$ (Perley, 2018).

$$V(\boldsymbol{b}) = R_C - iR_S = \iint I_\nu(\boldsymbol{s})e^{-2\pi i\boldsymbol{u}\cdot\boldsymbol{s}}\,d\Omega \tag{2.28}$$

[9]The spatial integral is in units of solid angle Ω, as the usual axes of the sky are given in arcseconds, or arcminutes.

If the baselines lie on a flat plane, the complex visibility is then a 2D Fourier transform of the sky brightness at the observing frequency, allowing an inverse transform to be performed to reconstruct the sky given complete sampling. The baseline vectors $\boldsymbol{u} = (u, v, w)$ are defined in units of wavelength, with the flat uv-plane defined as a special case having $w = 0$. In 3D coordinates, the unit vector $\hat{\boldsymbol{u}}$ is aligned East-West, $\hat{\boldsymbol{v}}$ is North-South, and $\hat{\boldsymbol{w}}$ points towards the source (or phase centre). If the antenna array is not coplanar to the source, baselines may vary in w, which invalidates the 2D Fourier transform. To solve this, many imaging packages project the visibilities onto the uv-plane (called w-projection) (Perley, 2018).

An individual antenna pair only samples a single point in $\boldsymbol{u}$, so to maximise coverage of the uv-plane, interferometer arrays use N antennas in tandem to achieve $N(N-1)$ unique pairs simultaneously. Similarly, the rotation of the Earth allows these baselines to move with respect to the source, sweeping through the uv-plane in an ellipse to increase coverage. As source declination approaches 0°, the ellipse flattens due to the source alignment with the Earth's rotation, reducing effective coverage. The final coverage of the uv-plane then depends on the arrangement and number of antennas, the location in the sky that is being observed, and the length of time that the source is being observed.

The incomplete coverage of the uv-plane results in a synthesised beam with radial spikes and periodics in image space, called the point spread function (PSF). The inverse transform of the sampled complex visibilities is then called the 'dirty' image, having the true sky convolved with the PSF. Several methods have been developed to deconvolve the PSF, with the most common method being the CLEAN algorithm (Högbom, 1974; Clark, 1980). This follows an iterative major-minor cycle to build a model of the sky brightness. Minor cycles operate in the image space, identifying source components in the residual image and adding to the sky model. The major cycle transforms the model image into visibilities and subtracts them from the observed visibilities. These residuals are then Fourier transformed to update the residual image. With tuned parameterisation, the CLEAN algorithm can successfully deconvolve the PSF and return an accurate image of the sky brightness.

In practice, noise and calibration errors add further complexity to the observed complex visibilities. These can be delay errors, antenna-dependent gain and phase variation, atmospheric effects, and instrumental noise. This can be summarised via the Measurement Equation, defining the observed visibilities $\boldsymbol{V}_{ij}$ between antennas i and j in terms of the ideal visibilities $\boldsymbol{V}_{ij}^{IDEAL}$ and the accumulation of corrupting effects for that baseline J_{ij} (Hamaker et al., 1996; Sault et al., 1996).

$$\boldsymbol{V}_{ij} = \boldsymbol{J}_{ij} \boldsymbol{V}_{ij}^{IDEAL} \tag{2.29}$$

The factor $\boldsymbol{J}_{ij}$ can be factored into a sequence of individual effects applied to the incoming wave, with different physical origins and importances. The specific calibration and imaging procedures applied to the data analysed in this thesis are presented in the following chapter.

Chapter 3

Calibration and Imaging

The observations to be calibrated are VLA X-band (8-12 GHz) data taken in C and D configurations of 12 galaxy cluster minihalos. The VLA antennas can be arranged in four standard configurations, A through D. A has the longest baselines and best angular resolution, trading off brightness sensitivity. The baseline length decreases towards D configuration, which has the poorest angular resolution, but highest brightness sensitivity. Due to the faint and extended nature of minihalos, the most suitable arrangement is the highly sensitive D configuration. The better angular resolution C configuration observations are added to allow accurate imaging the compact AGN sources.

The C configuration observations were performed between January and February of 2024, with project code 24A-411, and the D configuration observations were taken during February and March 2025 under project code 25A-157. The frequency range for X-band is 7.977-12.025 GHz with a total bandwidth of 4.048 GHz. This is split into 32 spectral windows with individual bandwidths of 128 MHz. Each spectral window has a spectral resolution of 2 MHz, resulting in a total of 64 channels. An overview of the targeted clusters is given in Table 3.1, while details of the radio observations of each cluster is given in Table **??**.

3.1 Our observations

- We are using VLA data in X-band, what does this mean and what are the specs for the VLA.
- These observations are of list minihalos and this setup, observation proposal.
- Original source of these minihalos is the G14 paper (among others).
- Table of minihalos + observing setup.

Cluster	R.A.$_{J2000}$ (h, m, s)	Decl$_{J2000}$ (deg, arcmin, arcsec)	z	Scale (kpc arcsec^{-1})
2A0335+096	03 38 40.5	+09 58 12.4	0.036	0.716
A2626	23 36 30.5	+21 08 47.6	0.055	1.069
A1795	13 48 52.5	+26 35 34.0	0.062	1.195
A478	04 13 25.3	+10 27 55.0	0.086	1.612
A1413	11 55 18.0	+23 24 19.0	0.141	2.482
A2204	16 32 47.0	+05 34 35.0	0.152	2.643
RXJ1720+2638	17 20 10.0	+26 37 32.0	0.160	2.758
RXJ2129+0005	21 29 39.9	+00 05 21.3	0.234	3.722
MS1455+2232	14 57 15.1	+22 20 34.0	0.257	3.990
Z3146	10 23 39.6	+04 11 11.0	0.288	4.329
RXCJ1115+0129[a]	11 15 52.0	+01 29 54.0	0.350	4.940
ACTJ0022-0036[b]	00 22 13.0	−00 36 33.0	0.804	7.521

Notes. Redshifts are from the NASA/IPAC Extragalactic Database (NED). Values for kpc arcsec^{-1} are calculated using the Cosmology Calculator I (Wright, 2006).
[a] Also known as ZwCl 1454.8+2233.
[b] Also known as ZwCl 1113.8+0144.

Table 3.1: List of observed clusters, sorted by redshift.

Table 3.2: Summary of the VLA X-band observations and calibration sources used.

Cluster	VLA Config.	Date	Time (min)	Primary Calib.	Secondary Calib.
2A0335+096	C	2024-01-27	45.3	3C48	J0321+1221
	D	2025-02-28	68.5	3C48	J0321+1221
A478	C	2024-01-27	45.3	3C48	J0424+0204
	D	2025-02-27	124.5	3C48	J0424+0204
A2204	C	2024-02-03	64	3C286	J1616+0459
MS1455+2232	C	2024-02-03	61	3C286	J1436+2321
RXJ1720+2638	C	2024-02-03	62.3	3C286	J1719+1745
A2204	C	2024-02-03	64	3C286	J0424+0204
A1413	C	2024-02-04	36.9	3C286	J1150+2417
	D	2024-02-28	129.2	3C286	J1150+2417
A1795	C	2024-02-04	71.5	3C286	3C286
RXCJ1115+0129	C	2024-02-04	73.1	3C286	J1058+0133
	D	2025-02-27	36.5	3C286	J1058+0133
Z3146	C	2024-02-04	73.1	3C286	J1041+0610
	D	2025-02-27	36.5	3C286	J1041+0610
A2626	C	2024-02-11	41.5	3C48	J2253+1608
ACTJ0022-0036	C	2024-02-11	109.5	3C48	J0022+0014
RXJ2129+0005	C	2024-02-11	138.1	3C48	J2136+0041

Notes.

3.2 How to calibrate a dataset

- We have the VLA calibration pipeline to remove stress from our lives.
- We have a non-neglible amount of RFI to add stress to our lives.
- A variety of steps but first pain, what is RFI and what does it cause.

3.3 Flagging data

- Initial testing showed that the pipeline flagging steps didn't work well for our data.
- AOFLAGGER worked much better and has a mostly out-of-box VLA script.
- How does AOFLAGGER work
- Integrating into a pipeline step to still use the pipeline

3.4 Calibration pipeline

- Work through each calibration step explaining what it does and why it is important.
- Final preparation for imaging is averaging down to further reduce noise.
- Notes about general trends from the calibration, with RFI in 10-12 GHz D config being bad, much of it being flagged.

3.5 Special mentions

- Either here or during following imaging section talk about the 2-3 datasets we processed slightly differently due to bad fits.
- Describe the fitting process for the custom bandpass.

3.6 Imaging

- When we make a raw image from the data, we still have a lot of artefacts from the point spread function, which comes from the beam.
- This means we need a more advanced imaging process to remove this to get a clean image. Implemented in a few different ways, we choose tclean in CASA.

- Tclean works by modelling the PSF, creating an empty model, find brightest point in residuals, model that as real, subtract model from residuals, repeat 20K times.
- This works mostly well, but we can improve results using self calibration.

3.6.1 Self calibration

- The phase calibration is the most iffy part for the main calibration, so we can make a test image and check the model seems reasonable.
- Then use that model to make small phase corrections to our dataset. Then repeat that a few times and these corrections should converge since it is circular.
- This is the final calibration steps that we do, phase corrections 20 degrees $\rightarrow$ 1 degree after a few rounds.

3.6.2 Final cleaning

- Between self calibrations this we also check the datasets and do some final manual cleaning.
- Clipping the datasets to a sensible amp range, removing spikes, SPWs where there is small amounts of data left.
- Overview of the `assess_spw` function used for this.

Chapter 4

How to Analysis

This chapter starts with an overview on imaging and how that is performed. Then we talk a bit about how we actually want to measure the flux of the minihalos. Could combine with actual results.

4.1 Final datasets

- Could have a table here talking about each minihalo, how much data is remaining, spws that are flagged out, some sort of visual.
- Maybe bar graph of spws, line or point graph for minihalo C/D (either total or differentiated by style) normalised to [0, 1]

4.2 Calibration Error

- Calibration error exists, and sources can vary in their flux amount.
- We can try and constrain the error and variance by comparing the secondary calibrator models for D config and the point source fluxes for the images.
- Fluxes for secondaries can vary a lot over a year (100%?) maybe, but lower per month. Can then compare to differences in sources in the image.
- If a trend is observed, likely calibration error. If no trend or close models, constrain calibration error and can see the variance in AGN etc.
- Can retrieve the point source flux by cutting out diffuse sources (uvrange) and by masking to a tight border.

4.3 Calculating the minihalo flux

- We can do this in a variety of ways, but we don't want the AGN contamination so we should subtract this out.
- Do this per dataset to account for variation over time, incorrect subtractions can result in negative bowls in emission, or remaining flux.
- Other problematic sources are dealt with the same way.
- Image the subtracted data and then we create a contour for significant emission (3 sigma) from the residual RMS.
- Calculate total flux in the region.
- Then we calculate the error using this formula from G14 modified for the AGN error since we think its lower.

Chapter 5

Results

- Sections for each minihalo with a brief explanation of imaging process, custom things we had to do to get results.
- Number of self calibration, sources subtracted, difficulties in imaging certain aspects.
- Discussion on the calibration error and variance in the observed point sources.
- Includes an image of the uvrange data showing point sources, image of the minihalo with contours, and anything else we desire.
- Includes discussion compared to previous results or measured values (likely G14) and any caveats or concerns.

Chapter 6

Discussion

- General findings from the above
- Any downsides caveats for the whole dataset
- Compare general results to things like G14 / G19 and how comparable these values are. Both literally and if valid comparison.

Chapter 7

Conclusion

- I'm sure I will have some conclusions here.

Bibliography

Ackermann, M., Ajello, M., Allafort, A., Baldini, L., Ballet, J., Barbiellini, G., Bastieri, D., Bechtol, K., Bellazzini, R., Blandford, R. D., Blasi, P., Bloom, E. D., Bonamente, E., Borgland, A. W., Bouvier, A., Brandt, T. J., Bregeon, J., Brigida, M., Bruel, P., Buehler, R., Buson, S., Caliandro, G. A., Cameron, R. A., Caraveo, P. A., Carrigan, S., Casandjian, J. M., Cavazzuti, E., Cecchi, C., Çelik, Ö., Charles, E., Chekhtman, A., Cheung, C. C., Chiang, J., Ciprini, S., Claus, R., Cohen-Tanugi, J., Colafrancesco, S., Cominsky, L. R., Conrad, J., Dermer, C. D., de Palma, F., Silva, E. d. C. e., Drell, P. S., Dubois, R., Dumora, D., Edmonds, Y., Farnier, C., Favuzzi, C., Frailis, M., Fukazawa, Y., Funk, S., Fusco, P., Gargano, F., Gasparrini, D., Gehrels, N., Germani, S., Giglietto, N., Giordano, F., Giroletti, M., Glanzman, T., Godfrey, G., Grenier, I. A., Grondin, M.-H., Guiriec, S., Hadasch, D., Harding, A. K., Hayashida, M., Hays, E., Horan, D., Hughes, R. E., Jeltema, T. E., Jóhannesson, G., Johnson, A. S., Johnson, T. J., Johnson, W. N., Kamae, T., Katagiri, H., Kataoka, J., Kerr, M., Knödlseder, J., Kuss, M., Lande, J., Latronico, L., Lee, S.-H., Lemoine-Goumard, M., Longo, F., Loparco, F., Lott, B., Lovellette, M. N., Lubrano, P., Madejski, G. M., Makeev, A., Mazziotta, M. N., Michelson, P. F., Mitthumsiri, W., Mizuno, T., Moiseev, A. A., Monte, C., Monzani, M. E., Morselli, A., Moskalenko, I. V., Murgia, S., Naumann-Godo, M., Nolan, P. L., Norris, J. P., Nuss, E., Ohsugi, T., Omodei, N., Orlando, E., Ormes, J. F., Ozaki, M., Paneque, D., Panetta, J. H., Pepe, M., Pesce-Rollins, M., Petrosian, V., Pfrommer, C., Piron, F., Porter, T. A., Profumo, S., Rainò, S., Rando, R., Razzano, M., Reimer, A., Reimer, O., Reposeur, T., Ripken, J., Ritz, S., Rodriguez, A. Y., Romani, R. W., Roth, M., Sadrozinski, H. F.-W., Sander, A., Saz Parkinson, P. M., Scargle, J. D., Sgrò, C., Siskind, E. J., Smith, P. D., Spandre, G., Spinelli, P., Starck, J.-L., Stawarz, Ł., Strickman, M. S., Strong, A. W., Suson, D. J., Tajima, H., Takahashi, H., Takahashi, T., Tanaka, T., Thayer, J. B., Thayer, J. G., Tibaldo, L., Tibolla, O., Torres, D. F., Tosti, G., Tramacere, A., Uchiyama, Y., Usher, T. L., Vandenbroucke, J., Vasileiou, V., Vilchez, N., Vitale, V., Waite, A. P., Wang, P., Winer, B. L., Wood, K. S., Yang, Z., Ylinen, T., and Ziegler, M. (2010). GeV Gamma-ray Flux Upper Limits from Clusters of Galaxies. *The Astrophysical Journal*, 717(1):L71–L78.

Bagchi, J., Enßlin, T. A., Miniati, F., Stalin, C. S., Singh, M., Raychaudhury, S., and Humeshkar,

N. B. (2002). Evidence for shock acceleration and intergalactic magnetic fields in a large-scale filament of galaxies ZwCl 2341.1+0000. *New Astronomy*, 7(5):249–277.

Blumenthal, G. R., Faber, S. M., Primack, J. R., and Rees, M. J. (1984). Formation of galaxies and large-scale structure with cold dark matter. *Nature*, 311:517–525.

Böhringer, H., Chon, G., Trümper, J., Kraan-Korteweg, R. C., and Schartel, N. (2025). Unveiling the largest structures in the nearby Universe: Discovery of the Quipu superstructure. *Astronomy & Astrophysics*, 695:A59.

Bond, J. R., Kofman, L., and Pogosyan, D. (1996). How filaments of galaxies are woven into the cosmic web. *Nature*, 380(6575):603–606.

Brunetti, G., Blasi, P., Reimer, O., Rudnick, L., Bonafede, A., and Brown, S. (2012). Probing the origin of giant radio haloes through radio and γ-ray data: the case of the Coma cluster. *Monthly Notices of the Royal Astronomical Society*, 426(2):956–968.

Brunetti, G. and Jones, T. W. (2014). Cosmic rays in galaxy clusters and their nonthermal emission. *International Journal of Modern Physics D*, 23(04):1430007.

Brunetti, G. and Lazarian, A. (2007). Compressible turbulence in galaxy clusters: physics and stochastic particle re-acceleration. *Monthly Notices of the Royal Astronomical Society*, 378(1):245–275.

Brunetti, G., Setti, G., Feretti, L., and Giovannini, G. (2001). Particle reacceleration in the Coma cluster: radio properties and hard X-ray emission. *Monthly Notices of the Royal Astronomical Society*, 320(3):365–378.

Brunetti, G., Zimmer, S., and Zandanel, F. (2017). Relativistic protons in the coma galaxy cluster: first gamma-ray constraints ever on turbulent reacceleration. *Monthly Notices of the Royal Astronomical Society*, 472(2):1506–1525.

Clark, B. G. (1980). An efficient implementation of the algorithm 'CLEAN'. *Astronomy & Astrophysics*, 89(3):377.

Dennison, B. (1980). Formation of radio halos in clusters of galaxies from cosmic-ray protons. *The Astrophysical Journal*, 239:L93–L96.

Dolag, K. and Enßlin, T. A. (2000). Radio halos of galaxy clusters from hadronic secondary electron injection in realistic magnetic field configurations. *Astronomy & Astrophysics*, 362:151–157.

Fabian, A. C. (2012). Observational Evidence of Active Galactic Nuclei Feedback. *Annual Review of Astronomy and Astrophysics*, 50:455–489.

Feretti, L., Giovannini, G., Govoni, F., and Murgia, M. (2012). Clusters of galaxies: observational properties of the diffuse radio emission. *The Astronomy and Astrophysics Review*, 20:54.

Fujita, Y., Matsumoto, T., and Wada, K. (2004). Strong Turbulence in the Cool Cores of Galaxy Clusters: Can Tsunamis Solve the Cooling Flow Problem? *The Astrophysical Journal*, 612(1):L9–L12.

Giacintucci, S., Markevitch, M., Brunetti, G., ZuHone, J. A., Venturi, T., Mazzotta, P., and Bourdin, H. (2014a). Mapping the Particle Acceleration in the Cool Core of the Galaxy Cluster RX J1720.1+2638. *The Astrophysical Journal*, 795(1):73.

Giacintucci, S., Markevitch, M., Cassano, R., Venturi, T., Clarke, T. E., and Brunetti, G. (2017). Occurrence of Radio Minihalos in a Mass-limited Sample of Galaxy Clusters. *The Astrophysical Journal*, 841(2):71.

Giacintucci, S., Markevitch, M., Cassano, R., Venturi, T., Clarke, T. E., Kale, R., and Cuciti, V. (2019). Expanding the Sample of Radio Minihalos in Galaxy Clusters. *The Astrophysical Journal*, 880(2):70.

Giacintucci, S., Markevitch, M., Venturi, T., Clarke, T. E., Cassano, R., and Mazzotta, P. (2014b). New Detections of Radio Minihalos in Cool Cores of Galaxy Clusters. *The Astrophysical Journal*, 781(1):9.

Giacintucci, S., Venturi, T., Markevitch, M., Brunetti, G., Clarke, T. E., and Kale, R. (2024). Are radio minihalos confined by cold fronts in galaxy clusters? minihalos and large-scale sloshing in a3444 and ms 1455.0+2232. *The Astrophysical Journal*, 961(1):133.

Gitti, M., Brunetti, G., Cassano, R., and Ettori, S. (2018). Radio-continuum surveys with SKA and LOFAR: a first look at the perspectives for radio mini-halos. *Astronomy & Astrophysics*, 617:A11.

Gitti, M., Brunetti, G., and Setti, G. (2002). Modeling the interaction between ICM and relativistic plasma in cooling flows: The case of the Perseus cluster. *Astronomy & Astrophysics*, 386:456–463.

Govoni, F. and Feretti, L. (2004). Magnetic fields in clusters of galaxies. *International Journal of Modern Physics D*, 13(08):1549–1594.

Grant, C. E., Bautz, M. W., Plucinsky, P. P., and Ford, P. G. (2024). The advanced ccd imaging spectrometer on the chandra x-ray observatory: twenty-five years of on-orbit operation.

Hamaker, J. P., Bregman, J. D., and Sault, R. J. (1996). Understanding radio polarimetry. I. Mathematical foundations. *Astronomy and Astrophysics Supplement*, 117:137–147.

HLavacek-Larrondo, J., Timmerman, R., Pfrommer, C., Osinga, E., Tevlin, L., Webb, T. M. A., Martorella, N., Zhang, X., van Weeren, R., Choi, H., Di Gennaro, G., Gendron-Marsolais, M.-L., and Rhea, C. (2025). Discovery of Diffuse Radio Emission in a Massive z = 1.709 Cool Core Cluster: A Candidate Radio Minihalo. *The Astrophysical Journal*, 987(2):L40.

Högbom, J. A. (1974). Aperture Synthesis with a Non-Regular Distribution of Interferometer Baselines. *Astronomy and Astrophysics Supplement*, 15:417.

Hudson, D. S., Mittal, R., Reiprich, T. H., Nulsen, P. E. J., Andernach, H., and Sarazin, C. L. (2010). What is a cool-core cluster? a detailed analysis of the cores of the X-ray flux-limited HIFLUGCS cluster sample. *Astronomy & Astrophysics*, 513:A37.

Ignesti, A., Brunetti, G., Gitti, M., and Giacintucci, S. (2020). Radio and X-ray connection in radio mini-halos: Implications for hadronic models. *Astronomy & Astrophysics*, 640:A37.

Jaffe, W. J. and Perola, G. C. (1973). Dynamical Models of Tailed Radio Sources in Clusters of Galaxies. *Astronomy & Astrophysics*, 26:423.

Kardashev, N. S. (1962). Nonstationarity of Spectra of Young Sources of Nonthermal Radio Emission. *Soviet Astronomy*, 6:317.

Keshet, U. and Loeb, A. (2010). Using radio halos and minihalos to measure the distributions of magnetic fields and cosmic rays in galaxy clusters. *The Astrophysical Journal*, 722(1):737.

Knowles, K., Baker, A. J., Bond, J. R., Gallardo, P. A., Gupta, N., Hilton, M., Hughes, J. P., Intema, H., López-Caraballo, C. H., Moodley, K., Schmitt, B. L., Sievers, J., Sifón, C., and Wollack, E. (2019). Gmrt 610 mhz observations of galaxy clusters in the act equatorial sample. *Monthly Notices of the Royal Astronomical Society*, 486(1):1332–1349.

Komissarov, S. S. and Gubanov, A. G. (1994). Relic radio galaxies: evolution of synchrotron spectrum. *Astronomy & Astrophysics*, 285:27–43.

Kravtsov, A. V. and Borgani, S. (2012). Formation of Galaxy Clusters. *Annual Review of Astronomy and Astrophysics*, 50:353–409.

Mantz, A. B., Allen, S. W., Morris, R. G., and Schmidt, R. W. (2016). Cosmology and astrophysics from relaxed galaxy clusters - iii. thermodynamic profiles and scaling relations. *Monthly Notices of the Royal Astronomical Society*, 456(4):4020–4039.

Markevitch, M. and Vikhlinin, A. (2007). Shocks and cold fronts in galaxy clusters. *Physics Reports*, 443(1):1–53.

Mazzotta, P. and Giacintucci, S. (2008). Do Radio Core-Halos and Cold Fronts in Non-Major-Merging Clusters Originate from the Same Gas Sloshing? *The Astrophysical Journal*, 675(1):L9.

Meekins, J. F., Fritz, G., Chubb, T. A., and Friedman, H. (1971). Physical Sciences: X-rays from the Coma Cluster of Galaxies. *Nature*, 231(5298):107–108.

Pacholczyk, A. G. (1970). *Radio astrophysics. Nonthermal processes in galactic and extragalactic sources*.

Perley, R. A. (2018). Fundamentals of radio interferometry. 16th Synthesis Imaging Workshop (talk slides). Presented at 16th Synthesis Imaging Workshop, 16–23 May 2018, National Radio Astronomy Observatory.

Perrott, Y. C., Carvalho, P., Elwood, P. J., Grainge, K. J. B., Green, D. A., Javid, K., Jin, T. Z., Rumsey, C., and Saunders, R. D. E. (2021). A 15.5 ghz detection of the galaxy cluster minihalo in rxj1720.1+2638. *Monthly Notices of the Royal Astronomical Society*, 508(2):2862–2880.

Peterson, J. R. and Fabian, A. C. (2006). X-ray spectroscopy of cooling clusters. *Physics Reports*, 427(1):1–39.

Pimbblet, K. A. (2005). A new algorithm for the detection of intercluster galaxy filaments using galaxy orientation alignments. *Monthly Notices of the Royal Astronomical Society*, 358(1):256–262.

Pratt, G. W., Arnaud, M., Biviano, A., Eckert, D., Ettori, S., Nagai, D., Okabe, N., and Reiprich, T. H. (2019). The Galaxy Cluster Mass Scale and Its Impact on Cosmological Constraints from the Cluster Population. *Space Science Reviews*, 215(2):25.

Press, W. H. and Schechter, P. (1974). Formation of Galaxies and Clusters of Galaxies by Self-Similar Gravitational Condensation. *The Astrophysical Journal*, 187:425–438.

Reimer, O., Pohl, M., Sreekumar, P., and Mattox, J. R. (2003). EGRET Upper Limits on the High-Energy Gamma-Ray Emission of Galaxy Clusters. *The Astrophysical Journal*, 588(1):155–164.

Riseley, C. J., Biava, N., Lusetti, G., Bonafede, A., Bonnassieux, E., Botteon, A., Loi, F., Brunetti, G., Cassano, R., Osinga, E., Rajpurohit, K., Röttgering, H. J. A., Shimwell, T., Timmerman, R., and van Weeren, R. J. (2023). A meerkat-meets-lofar study of abell 1413: a moderately disturbed non-cool-core cluster hosting a 500 kpc 'mini'-halo. *Monthly Notices of the Royal Astronomical Society*, 524(4):6052–6070.

Riseley, C. J., Bonafede, A., Bruno, L., Botteon, A., Rossetti, M., Biava, N., Bonnassieux, E., Loi, F., Vernstrom, T., and Balboni, M. (2024). A “meerkat-meets-lofar” study of the complex multi-component (mini-)halo in the extreme sloshing cluster abell 2142. *Astronomy & Astrophysics*, 686:A44.

Riseley, C. J., Rajpurohit, K., Loi, F., Botteon, A., Timmerman, R., Biava, N., Bonafede, A., Bonnassieux, E., Brunetti, G., Enßlin, T., and et al. (2022). A MeerKAT-meets-LOFAR study of MS 1455.0 + 2232: a 590 kiloparsec ’mini’-halo in a sloshing cool-core cluster. *Monthly Notices of the Royal Astronomical Society*, 512(3):4210–4230.

Sarazin, C. L. (1999). The Energy Spectrum of Primary Cosmic-Ray Electrons in Clusters of Galaxies and Inverse Compton Emission. *The Astrophysical Journal*, 520(2):529–547.

Sarkar, A., Miller, E., Ota, N., Kilbourne, C., McNamara, B., Sun, M., Lovisari, L., Ettori, S., Eckert, D., Szymkowiak, A., and et al. (2025). XRISM reveals complex multi-temperature structures in the Abell 2029 galaxy cluster. *Publications of the Astronomical Society of Japan*, 77:S254–S269.

Sault, R. J., Hamaker, J. P., and Bregman, J. D. (1996). Understanding radio polarimetry. II. Instrumental calibration of an interferometer array. *Astronomy and Astrophysics Supplement*, 117:149–159.

Savini, F., Bonafede, A., Brüggen, M., van Weeren, R., Brunetti, G., Intema, H., Botteon, A., Shimwell, T., Wilber, A., Rafferty, D., Giacintucci, S., Cassano, R., Cuciti, V., de Gasperin, F., Röttgering, H., Hoeft, M., and White, G. (2018). First evidence of diffuse ultra-steep-spectrum radio emission surrounding the cool core of a cluster. *Monthly Notices of the Royal Astronomical Society*, 478(2):2234–2242.

Savini, F., Bonafede, A., Brüggen, M., Rafferty, D., Shimwell, T., Botteon, A., Brunetti, G., Intema, H., Wilber, A., Cassano, R., Vazza, F., van Weeren, R., Cuciti, V., De Gasperin, F., Röttgering, H., Sommer, M., Bîrzan, L., and Drabent, A. (2019). A lofar study of non-merging massive galaxy clusters. *Astronomy & Astrophysics*, 622:A24.

Schneider, P. (2014). *Extragalactic Astronomy and Cosmology*. Springer Berlin, Heidelberg.

Serlemitsos, P. J., Smith, B. W., Boldt, E. A., Holt, S. S., and Swank, J. H. (1977). X-radiation from clusters of galaxies: spectral evidence for a hot evolved gas. *The Astrophysical Journal*, 211:L63–L66.

Tudorache, M. N., Jung, S. L., Jarvis, M. J., Heywood, I., Ponomareva, A. A., Vărăşteanu, A. A., Maddox, N., Yasin, T., and Glowacki, M. (2025). A 15 mpc rotating galaxy filament at redshift z = 0.032. *Monthly Notices of the Royal Astronomical Society*, 544(4):4306–4316.

van Weeren, R. J., de Gasperin, F., Akamatsu, H., Brüggen, M., Feretti, L., Kang, H., Stroe, A., and Zandanel, F. (2019). Diffuse Radio Emission from Galaxy Clusters. *Space Science Reviews*, 215(1):16.

van Weeren, R. J., Intema, H. T., Lal, D. V., Andrade-Santos, F., Brüggen, M., de Gasperin, F., Forman, W. R., Hoeft, M., Jones, C., Nuza, S. E., Röttgering, H. J. A., and Stroe, A. (2014). A Distant Radio Mini-halo in the Phoenix Galaxy Cluster. *The Astrophysical Journal*, 786(2):L17.

Venturi, T., Rossetti, M., Brunetti, G., Farnsworth, D., Gastaldello, F., Giacintucci, S., Lal, D. V., Rudnick, L., Shimwell, T. W., Eckert, D., Molendi, S., and Owers, M. (2017). The two-component giant radio halo in the galaxy cluster abell 2142. *Astronomy & Astrophysics*, 603:A125.

Vikhlinin, A., Kravtsov, A., Forman, W., Jones, C., Markevitch, M., Murray, S. S., and Van Speybroeck, L. (2006). Chandra sample of nearby relaxed galaxy clusters: Mass, gas fraction, and mass-temperature relation. *The Astrophysical Journal*, 640(2):691.

Voit, G. M. (2005). Tracing cosmic evolution with clusters of galaxies. *Reviews of Modern Physics*, 77(1):207–258.

Wik, D. R., Hornstrup, A., Molendi, S., Madejski, G., Harrison, F. A., Zoglauer, A., Grefenstette, B. W., Gastaldello, F., Madsen, K. K., Westergaard, N. J., Ferreira, D. D. M., Kitaguchi, T., Pedersen, K., Boggs, S. E., Christensen, F. E., Craig, W. W., Hailey, C. J., Stern, D., and Zhang, W. W. (2014). NuSTAR Observations of the Bullet Cluster: Constraints on Inverse Compton Emission. *The Astrophysical Journal*, 792(1):48.

Wright, E. L. (2006). A Cosmology Calculator for the World Wide Web. *The Publications of the Astronomical Society of the Pacific*, 118(850):1711–1715.

XRISM Collaboration, Audard, M., Awaki, H., Ballhausen, R., Bamba, A., Behar, E., Boissay-Malaquin, R., Brenneman, L., Brown, G. V., Corrales, L., and et al. (2025). XRISM/Resolve view of Abell 2319: Turbulence, sloshing, and ICM dynamics. *Publications of the Astronomical Society of Japan*.

Zandanel, F. and Ando, S. (2014). Constraints on diffuse gamma-ray emission from structure formation processes in the Coma cluster. *Monthly Notices of the Royal Astronomical Society*, 440(1):663–671.

Zandanel, F., Pfrommer, C., and Prada, F. (2014). On the physics of radio haloes in galaxy clusters: scaling relations and luminosity functions. *Monthly Notices of the Royal Astronomical Society*, 438(1):124–144.

ZuHone, J. A., Brunetti, G., Giacintucci, S., and Markevitch, M. (2015). Testing Secondary Models for the Origin of Radio Mini-Halos in Galaxy Clusters. *The Astrophysical Journal*, 801(2):146.

ZuHone, J. A., Markevitch, M., Brunetti, G., and Giacintucci, S. (2013). Turbulence and Radio Mini-halos in the Sloshing Cores of Galaxy Clusters. *The Astrophysical Journal*, 762(2):78.